Aligning LLMs with Biomedical Knowledge using Balanced Fine-Tuning

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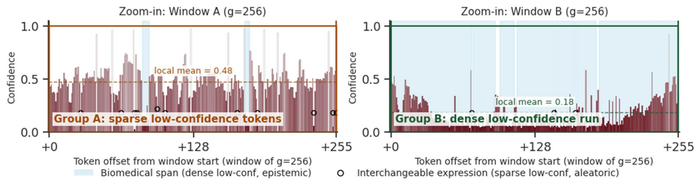
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NEURAL INFORMATION PROCESSING SYSTEMS

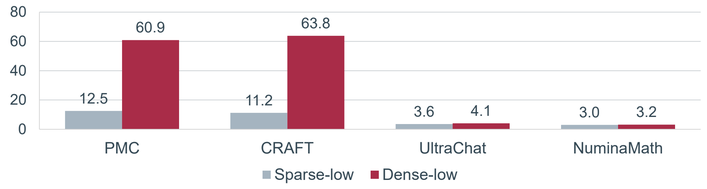
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01 Experimental observations

Teacher-forced PMC text reveals two regimes: isolated low-confidence tokens and contiguous difficult regions (g = 256).



Biomedical entity / expert-concept share (%)



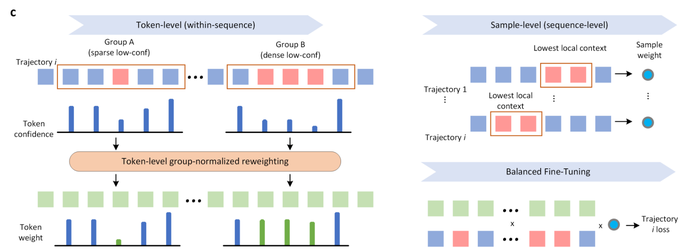
CRAFT uses expert concept spans. General-domain controls show little entity enrichment. These measurements support a knowledge-gap interpretation, rather than directly labeling epistemic uncertainty.

02 Motivation

DFT downweights every low-probability token. Biomedical learning needs to preserve difficult knowledge-bearing runs while controlling isolated outliers, then prioritize sequences containing hard regions.

03 Balanced Fine-Tuning

One local confidence signal allocates training budget within each response and across responses.



Within a sequence: local relative confidence

$$\rho_{\beta,t} = \mathrm{clip}(c_{\beta,t} / (C_{\beta,t}^{\mathrm{loc}} + \varepsilon), 0, 1)$$

For c = 0.10: $C^{\mathrm{loc}} = 0.87$ gives $\rho = 0.115$, while $C^{\mathrm{loc}} = 0.20$ gives $\rho = 0.50$.

Across sequences: weakest local region

$$s_\beta = \mathrm{sg}(1 - \min_t C_{\beta,t}^{\mathrm{loc}})$$

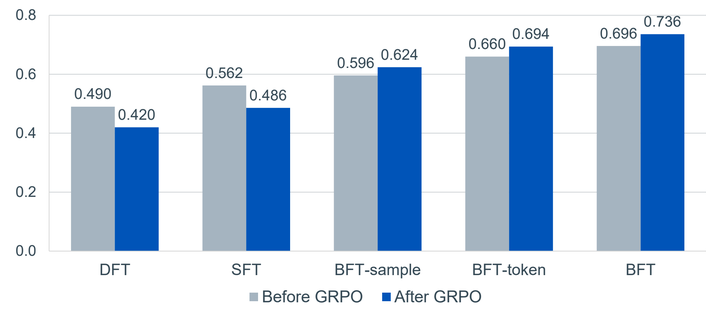
$$L = (1/B)\, \Sigma_\beta\, s_\beta\, [\Sigma_t\, m_{\beta,t}\, \mathrm{sg}(\rho_{\beta,t})\, \ell_{\beta,t}] \,/\, [\Sigma_t\, m_{\beta,t} + \varepsilon]$$

C^{loc}: centered-window mean of target probabilities. $\ell = -\log c$. m selects response tokens. Both weights use stop-gradient (sg). g = 256 is the only added tunable hyperparameter.

04 Experimental results

Chemical perturbation reasoning

70B backbones, joint Up/Down/No accuracy
GRPO on 45 cell lines, evaluation on 5 held-out lines



+13.4 pp over SFT before GRPO

BFT 0.696 vs SFT 0.562 under the shared recipe.
Token-only 0.660 and sample-only 0.596 are intermediate.

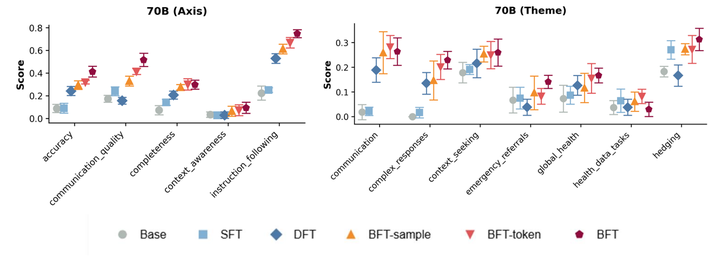
+4.0 pp after sparse-reward GRPO

All BFT variants improve. SFT loses 7.6 pp and DFT loses 7.0 pp under the same GRPO setup.

BFT vs Gemini-2.5-Flash: +0.018, 95% CI [−0.004, 0.040], p = 0.088. The difference is not statistically significant.

Medical & biological reasoning

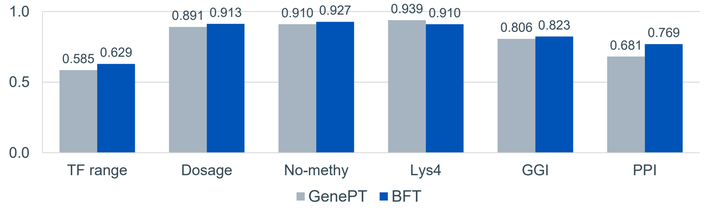
HealthBench Hard, 70B axis- and theme-wise scores



BFT improves more consistently across 14B / 32B / 70B. In GeneAgent, it achieves the best overall ROUGE among aligned open-weight models. Error bars denote decoding variability, not training-seed variability.

Gene & cell representations

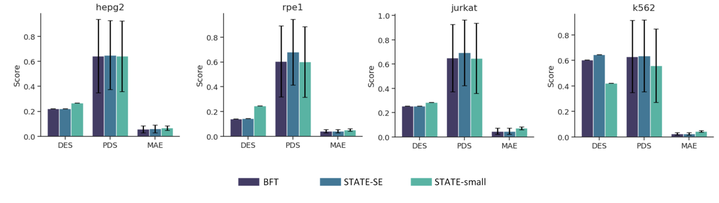
Gene-level AUC, matched ada-002 encoder



BFT cell-type clustering: ARI 0.34, AMI 0.58, ASW 0.061, best among evaluated methods.

Best average gene AUC: BFT 0.829 vs GenePT 0.802. GenePT remains higher on Lys4. BFT-generated profiles and NCBI summaries differ in source and construction.

Perturbation-response prediction



BFT profile embeddings perform comparably to standard STATE inputs across four datasets. Profile generation is zero-shot for these datasets. The STATE decoder remains trained.

arxiv.org/abs/2511.21075
github.com/TencentAILabHealthcare/BFT
shapsider.github.io/BFT